Git and Github Assignment.

Definition:

Git and Github are closely related tools used in software development, but they serve different roles in managing and collaborating on Source Code.

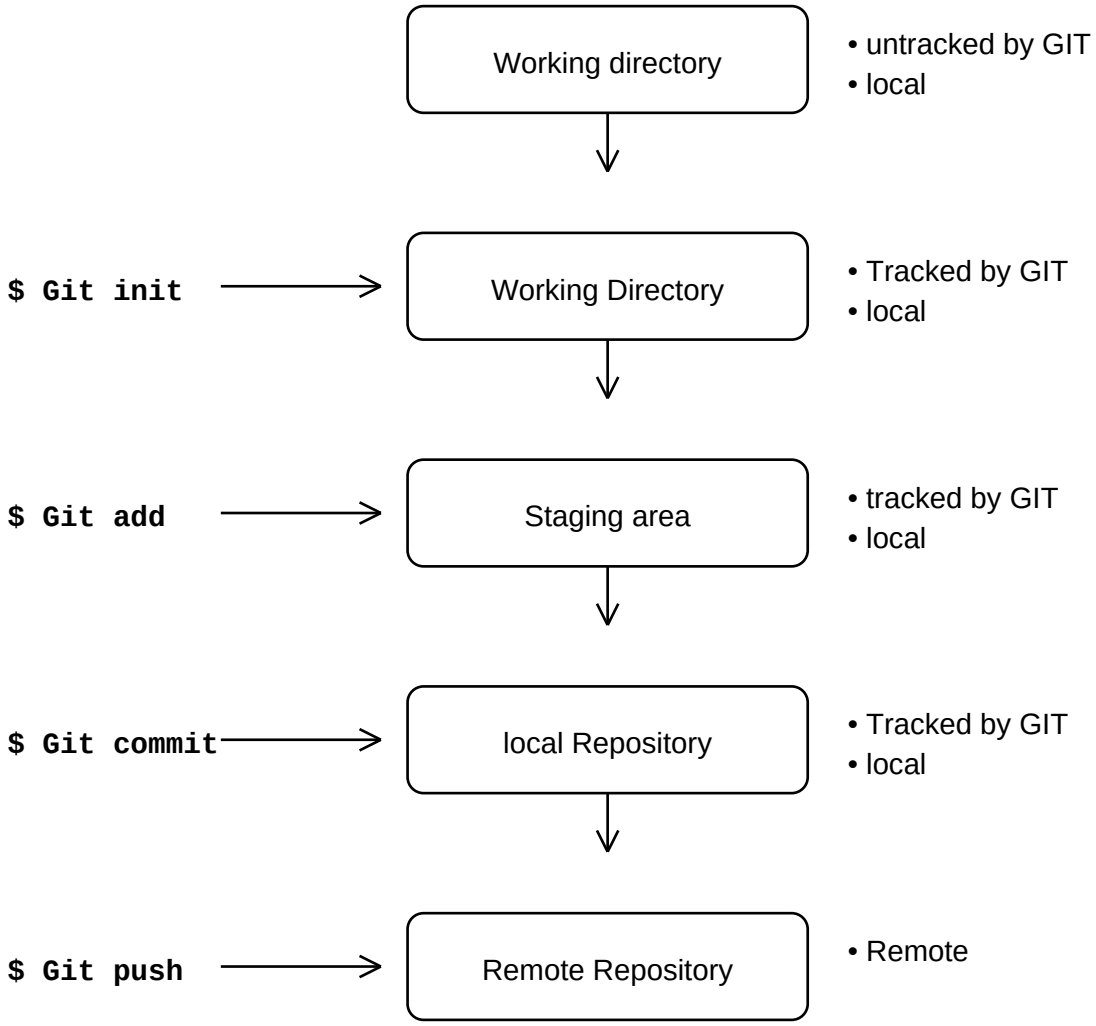
-> Git:

A distributed version control system used to track and manage code changes locally with features like branching and merging.

-> Github:

A web based platforms that hosts Git repositories and enables collaboration through pull requests, Issues and team management.

Git lifecycle:



Basic Git Commands:

1) Git version -> Checks the installed version of Git (Command: **`git --version`**)

2) **`git config --global user.name "Name"`**: it helps to set your identity for all commits.

3) **`git config --global user.email "mail"`**: it helps to set your email address (must match our github)

4) **`git init`** -> Initializes a new, empty git repository in the current folder.

5) **`ls`** -> to view the list of directories and files in the system.

6) **`git clone <url>`** -> Downloads an existing repository from a remote server (from github)

7) **`git status`**: It shows which files are modified, staged or untracked

8) **`git add <file>`**: Adds a specific file to the staging area

9) **`git add .`** -> Add all changed files in the current directory to staging.

10) **`git rm <file>`** -> Deletes a file from both the working directory and the Index.

11) **`touch, nano, vi`** -> to Create a files empty.

12) **`mkdir`** -> to Create a Folder/directories

13) **`cd`** -> to change the directory.

14) **`cat`** -> It is used to view the file content.

15) **`git log`** -> to view the changes that we made in the git

16) **`git log --oneline`** -> to view the changes in oneline/ Shorter Version.

17) **`git commit -m "Message"`**: to commit the changes with message to show other person to view and observe the change.

18) **`git commit -am "message"`**: Shortcut to git add . and git commit in One command (tracked file only)

19) **`git branch`** -> List all local branches.

20) **`git branch <name>`** -> Creates a new branch.

21) **`git switch <name>`** -> to switch the branch.

22) **`git switch -c <name>`** -> to Create a branch and switch to new branch created in one line command.

23) **`git remote add origin <url>`** Connects your local repo to remote server.

24) **`git remote -v`** -> list all remote repositories link to your local project.

25) **`git push -u origin main`** -> uploads your commits to the remote branch.